



Machine Vision-Assisted Welding Defect Detection System with Convolutional Neural Networks

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Abstract

This study explores the application of a machine vision system integrated with convolutional neural network (CNN) for detecting and classifying welding defects. By leveraging the power of deep learning approaches, the proposed approach aims to enhance the efficiency and reliability of defect classification. This method not only reduces human dependency, but also establishes a framework for automated welding quality control systems. A CNN-based machine vision system has been developed to classify welding defects in radiographic images. Particularly, two transfer learning algorithms, mainly, ResNet-18 and ResNet-50, have been applied and evaluated in order to determine the most effective method in detecting and classifying weld defects. The dataset covered three classes of weld defects: cracks, lack of penetration, and porosity. The performance of each ResNet-based CNN model was assessed using performance evaluation metrics and visualization techniques. ResNet-50 emerged as the best performing model and had a strongest response in the weld defects regions, achieving an average accuracy of 96.061%. This model proved effective in detecting and classifying defects, demonstrating its potential to significantly enhance the reliability and automation of detection and recognition.

Keywords Welding defect detection · Classification · Machine learning · Convolutional neural network (CNN) · Machine vision

1 Introduction

In recent years there has been a rapid development of manufacturing technology that focuses on automatic welding technology which is widely utilized in numerous industries including aerospace, nuclear, automotive, machine design, shipbuilding, construction, chemical, and petrochemical engineering [1]. Welding defects are a major concern in these industries as defects in the welded joints may compromise the structural performance result in catastrophic failures and incur significant costs. Therefore, detecting and

classifying these defects effectively is crucial to maintaining product safety and performance [2].

There are three traditional techniques for examining the weld quality: visual inspection, destructive testing, and non-destructive evaluation (NDE), while effective, but have inherent limitations. In particular, visual inspection can only pick up the surface defects [3]. Destructive testing such as tensile and bending testing are also effective, still, they destroy the tested material, generate waste, consume cost and time, and are limited with the sample size and real-world application. While NDE techniques (e.g. liquid penetrant, ultrasound, and X-ray) highlight the surface and sub-surface defects, but some are harmful or need skilled technicians [4, 5]. One of the widely used NDE techniques is the radiographic testing that can detect the internal welding defects by generating radiographic images identifying different types of defects include porosity, slag inclusions, cracks, undercut, burn-through, and lack of penetration [6]. However manual interpretation of radiographic images may be unreliable as it is susceptible to human error, subjectivity, and inconsistency highlighting the necessity for implementing advanced automated technologies [7]. Therefore,

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integrating machine vision systems and machine learning (ML) techniques has emerged as a promising alternative for automating defect detection [8]. Also, choosing the proper classifier is a crucial factor influencing the effectiveness of defect classification.

Many researchers focused on using different ML techniques to automatically detect weld defects [9]. Among ML methodologies, convolutional neural networks (CNNs), a deep learning tool, have shown exceptional performance in image classification, enabling high-precision detection and classification of welding defects [10–12]. In this context, Kanthalakshmi et al. [13] reported that the implementation of pure CNN successfully classified three classes of TIG welding defects dataset with an accuracy of 96.1%. Kumaresan et al. [14] used two convolutional neural network-based architectures; VGG16 and ResNet-50; trained by X-ray radiograph image dataset of welded samples. It was reported that image-based identification model based on the VGG16 neural network performed well with a higher average accuracy of 90%. Kuppuswamy et al. [15] applied two CNN models; ResNet-18 and ResNet-50; and compared their ability to monitor TIG welding defects and both models achieved an accuracy of 99%. Palma-Ramírez et al. [16] used a CNN model based on ResNet-50 architecture trained with three datasets; high-density RIAWELC images, GDXray, and a private dataset of low image quality; for classification four types of radiographic weld defects images. The classification performance of the used model had an accuracy of 98.75%, 90.255%, and 75.83% for RIAWELC, GDXray, and a private dataset of low image quality, respectively. Gavrilov et al. [17] applied CNN approach based on three architectures; VGG16, Alexnet, and Resnet-50, and compared their performance in classifying the images of welding seam defects. Resnet-50 in combination with Random Forest showed the highest classification accuracy of 97.76%. Khan et al. [18] evaluated and compared the performance of three CNN classifiers: EfficientNetB3, ResNet-50, and MobileNetV2, to determine the most effective model for surface defect detection of steel components. It was demonstrated that EfficientNetB3 model accurately detected the defects with a higher accuracy of 98.1%.

Other attempts were done to improve the defect recognition performance through incorporation of ML with the deep transfer learning approaches [19, 20]. For instance, Sizyakin et al. [21] utilized convolutional neural networks combined with support vector machine (SVM) methods to detect four defects' types of welded joints, which proved a higher classification performance compared with the use of a convolutional neural network alone. Liu et al. [22] reported that AlexNet network with SVM achieved a better performance in detecting welding defects than other related recognition algorithms (VGG16, GoogleNet, Mobilenetv2, ResNet-18, and InceptionV3) with an accuracy of 97.59%. Kumaresan

et al. [23] applied VGG16 and ResNet50 CNNs models combined with different ML algorithms (Logistic Regression (LR), Support Vector Machine (SVM), and Random Forest (RF)) in weld defect classification. The results showed that the ResNet-50 model combined with SVM classifier improved the classification performance with an accuracy of 98%. Also, Zhang et al. [24] proposed an image classification model based on improved ResNet model which is optimized by efficient channel attention (ECA) module. The results showed that the proposed ResNet-34-ECA model accurately classified the images of welding defects with accuracy reached 98.2%. Jun Sun et al. [25] proposed a modified background subtraction (MBS) method based on Gaussian mixture models, and compared its performance with single Gaussian (SG), mixture of Gaussians (MOG), kernel density estimation (KDE), and subspace learning using PCA (SLPCA). The results showed that MBS was the most suitable in weld defect detection among other four algorithms with an accuracy of 95%.

In the light of the literature survey, deep learning techniques such as CNNs, are receiving more attention in automated detection and classification of weld defects from radiography images to classify reliably welding defects more accurate than traditional techniques. Therefore, the motivation of the current study is to develop a model that can detect and classify defects with a comparatively high accuracy to increase product quality and save costs. This highlights the potential of deep learning in welding inspection methodologies. In this study, the application of CNN-based machine vision system has been explored for classifying welding defects in radiographic images. In particular, two transfer learning algorithms, namely ResNet-18 and ResNet-50, have been applied and evaluated in order to determine the most effective method in detecting and classifying weld defects.

2 Methodology

This study addresses three types of welding defects: crack, lack of penetration, and porosity; which can significantly compromise the strength and integrity of weld joints. Figure 1 presents examples of different types of welding defects radiographic images used in training the CNN classification models. Firstly, the X-ray images undergo multiple multi-image processing operations to lessen the noise effects and improve the radiographic images' quality. Secondly, the dataset of welding defect images has been directly fed into the CNN model, which automatically extracts features and classifies the defect type. To train the deep learning model the dataset has been divided into three subsets: 70% for training (504 images per class $\times$ 3 classes = 1,512 images), 15% for validation (110 images per class $\times$ 3 classes = 330 images), and 15% for testing (110 images per class $\times$ 3

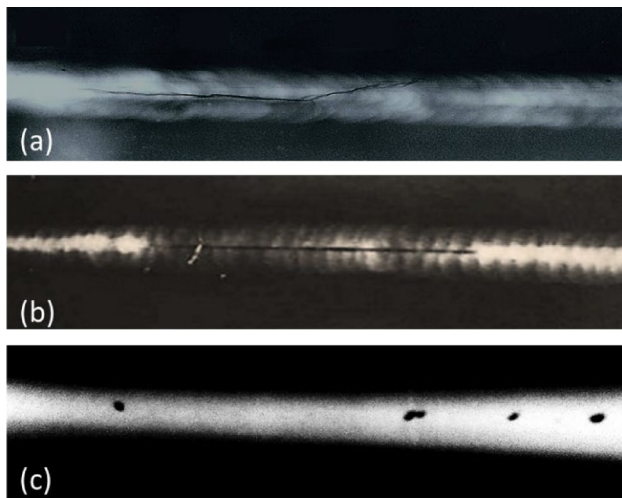


Fig. 1 Radiographic images of welding defect types used in training data set; **a** crack, **b** lack of penetration, and **c** porosity

classes = 330 images). The more parameters are trained in the model, the better and more accurate classification results are obtained. This study implemented two commonly used transfer learning classification benchmark architectures, including ResNet-18 and ResNet-50. The developed system was implemented using MATLAB software to detect the welding defect type based on the acquired radiographic images of the welding defect.

2.1 Image Preprocessing

Weld defect detection is affected by the radiography images quality; therefore, the preprocessing of the images is essential to enhance the quality and uniformity of the input data before further analysis [26]. First, all radiography images were cropped and resized to a fixed resolution using bilinear interpolation to standardize the input dimensions. This ensured consistency across the dataset and facilitated efficient processing by the neural network. Next, a Gaussian filter with a standard deviation of 2 was applied to each image to reduce noise while preserving important structural details. Gaussian filtering is well-suited for noise suppression in images because of its smooth, bell-shaped response, which attenuates high-frequency components that may represent noise, without significantly degrading the edges and textures. To further enhance the image contrast, particularly in regions with low local contrast, Adaptive Histogram Equalization (AHE) was employed. AHE dynamically adjusts the contrast of the image based on the intensity distribution of local regions, effectively enhancing contrast in both bright and dark areas. This technique is especially useful in handling non-uniform lighting conditions, which are common in real-world image acquisition. As a result, the enhanced

contrast facilitates better feature extraction by the subsequent image analysis algorithms.

2.2 Convolutional Neural Networks Approaches

Convolutional Neural Networks (CNNs) are type of deep learning techniques particularly well-suited for tasks involving spatial data, such as images and videos. They are widely used in computer vision applications, including image recognition, object detection, and semantic segmentation. CNNs are a class of deep learning models specifically designed to process data with a grid-like topology such as images. They have become the state-of-the-art approach for image classification tasks due to their ability to automatically learn hierarchical feature representations from raw pixel data. CNNs exploit local spatial correlations through convolutional layers effectively reducing the complexity of manual feature extraction [21]. As a result, the use of CNNs has been grew in several applications.

2.3 Proposed ResNet Models

The Residual Network (ResNet) architecture is renowned for its depth and effectiveness in image classification. It introduces the concept of residual learning, which enables training very deep networks by allowing gradients to flow through the network without vanishing or exploding. This characteristic is particularly beneficial for tasks like welding defect classification, where complex patterns need to be recognized [27, 28].

In this study, ResNet-18 and ResNet-50 were employed to classify welding defects, allowing for a comparative analysis of their performance. Each model was fine-tuned on the same dataset of welding images, ensuring consistency in training conditions and evaluation metrics. The comparative analysis of ResNet-18 and ResNet-50 in classifying welding defects not only highlights the strengths and weaknesses of each architecture, but also underscores the importance of utilizing robust measurement techniques. By employing these advanced deep learning methods and comprehensive evaluation metrics, this study aims to enhance the accuracy and reliability of welding defect classification, contributing to improved quality assurance and safety in welding practices. The architecture of the proposed ResNet-based CNN models is illustrated in Fig. 2.

2.4 Measurement Techniques for Performance Evaluation

The overall performance of each ResNet architecture applied in this study has been examined using a combination different evaluation criteria including both performance metrics and visualization techniques. This dual approach provides a

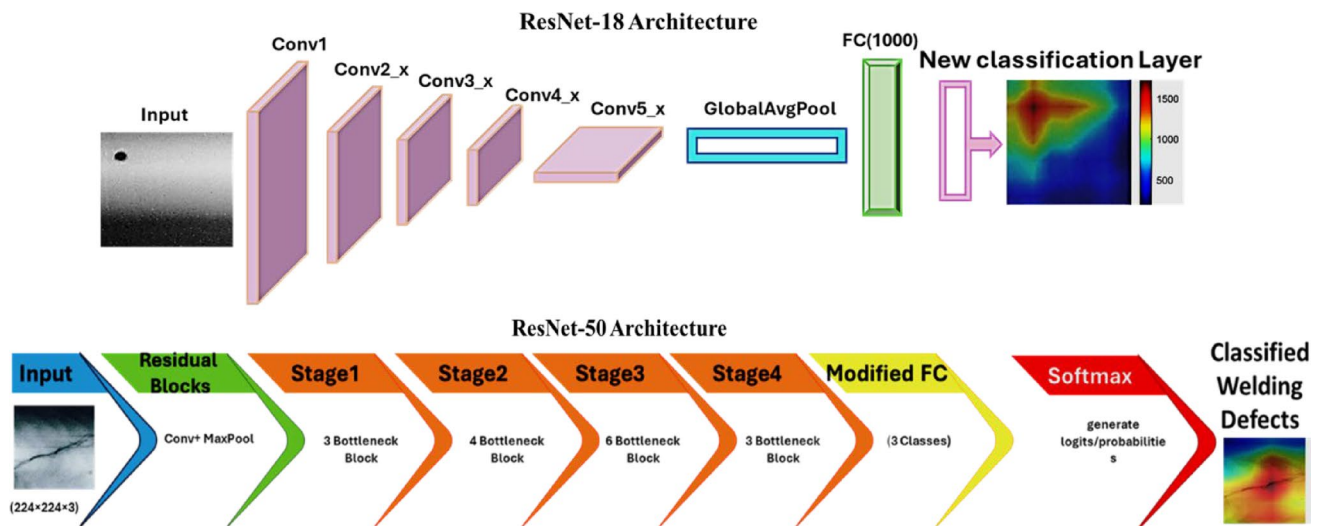


Fig. 2 The architecture of the used ResNet-based CNN models

comprehensive evaluation and understanding of the architectures' effectiveness in classifying welding defects and highlights areas for potential improvement.

2.4.1 Performance Metrics

Six common image classification performance metrics have been used as a quantitative approach, mainly, accuracy, precision, recall, F1-score, ROC curve, and AUC. These metrics are derived from the number of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) samples produced during testing of the CNN architectures. The definition and mathematical formulations are illustrated as follows [29–31].

Accuracy, is one of the most intuitive metrics to evaluate the model defined as the ratio of correctly predicted instances to the total instances, indicating the overall correctness of the CNN architecture's predictions, is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision and recall, precision reflects the accuracy of positive predictions (i.e., correctly identified defects), whereas recall measures the model's ability to identify all actual defects. This duality is crucial in assessing how well the models handle class imbalances, common in welding defect datasets where defective samples may be less frequent.

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

F1-Score, is a measure of model balance performance and serves as a harmonic mean of precision and recall, offering a single metric that balances the two. It is particularly valuable when dealing with uneven class distributions, ensuring that both false positives and false negatives are accounted for.

$$F1_{score} = 2 * \frac{precision * recall}{precision + recall}$$

ROC Curve and AUC, The Receiver Operating Characteristic (ROC) curve is a graphical representation used to plots the true positive rate (TPR) (sensitivity or recall) against the false positive rate (FPR) (specificity) at various threshold settings, allowing for an analysis of the trade-offs between sensitivity and specificity. A perfect classifier has a point at (0,1), indicating 100% TPR with 0% FPR. Meanwhile, the Area Under Curve (AUC) is a single scalar value summarizing the performance of the model by calculating the area under the ROC curve. It provides a measure of the model's ability to distinguish between positive and negative classes and quantifies this performance, as higher values indicating better model performance across all classes.

2.4.2 Visualization Techniques

Visualization techniques play a vital role in identifying the model classification performance as a qualitative approach to understand the data trends, relationships, and overall classification performance of the model. In order to visualize better understanding of classification distribution and depict prediction errors and class-wise performance, confusion

matrix visualization and class activation maps have been selected during this study as follows [23, 32, 33]:

Confusion Matrix, is a contingency table designed to compare two dimensions: the "actual" and the "predicted" classes and both dimensions containing the same sets of classes. This matrix provides a comprehensive evaluation of the model's classification performance across all defect categories, highlighting which defects were commonly confused with one another and insights into correct classifications and misclassifications.

Grad-CAM (Gradient-weighted Class Activation Mapping), highlights the regions in an image that contributed most to the model's decision. By understanding which features the model focuses on, practitioners can gain insights into the reliability of the model's predictions and improve interpretability.

3 Results and Discussions

The goal of network training is to produce a network that has successfully extracted the defining features for each weld feature and to use the convolutional filters to correlate these features with any given image and predict which weld feature it belongs to. A deeper network was able to extract more complex features but also runs the risk of overfitting to the training data. In order to find out how deep the network is required to be, the performance of the ResNet-50 model is compared to a shallower ResNet-18 model.

During the training process, the learning performance of each model was monitored by using the learning curves. Figure 3 shows the trend for accuracy and objective function loss value during the training and validation of the used CNN classification models to understand the model's learning progress and generalization ability. The training learning curves appear satisfactory when increasing training accuracy and decreasing training loss. From the figure, it can be observed that balanced learning achieved as the number of epochs and iterations increase. After 20 epochs, 940 iterations, and initial learning rate of 0.0001, the examined CNN classification models achieve the maximum accuracy and the loss value of objective function progressively smooths out, demonstrating the CNN's convergence. As shown in Fig. 3a, the training process of ResNet-18 model has an accuracy of 98.79% running in single CPU mode for a period of 161 min 36 s. While ResNet-50 model has an accuracy of 98.48% running for a period of 329 min 34 s as shown in Fig. 3b. After training the model, the test data set is used to check the classification accuracy and error of the proposed model. For our proposed CNN classification models, the average testing accuracy is 92.727% and 96.061% for ResNet-18 and ResNet-50, respectively. That implies that

our proposed CNN classification model achieves perfect training performance.

To facilitate a quantitative comparison, the effectiveness of each CNN architecture in the welding defect classification task has been assessed using the five-performance metrics previously mentioned, namely precision, recall, F1 score, ROC, and AUC. The results of these performance indices are detailed in Table 1 which illustrates performance comparison for the used pretrained CNN models. In general, the selected CNN models achieve high classification performance, however, ResNet-50 architecture is more effective in correctly identifying welding defect class.

Specifically, ResNet-50 exhibits the highest precision and recall in identifying the defective instances (100% and 90.9% for crack and 96.49% and 100% for lack of penetration defects, respectively). While ResNet-18 shows the lowest precision and recall (96.8% and 82.72% for crack and 86.77% and 95.45% for lack of penetration, respectively). Precision and recall, however, ResNet-18 leading at 95.65% and 100%, respectively, suggesting its higher accuracy in classifying porosity defects. F1-score is high when both the precision and recall rates are high. Thus, ResNet-50 achieve the highest F1-scores with 95.23% and 98.21% in detecting crack and lack of penetration defects, respectively, which reflect its overall performance robustness in classifying the radiographic images of welding defects. Conversely, ResNet-18 exhibits lower F1-scores in identifying crack and lack of penetration defects (89.21% and 90.9%), but highest value with 97.77% in porosity defect detection. The improved performance can be attributed to ResNet-50's deeper architecture (50 layers), which allows it to learn more abstract and complex features than the shallower ResNet-18 (18 layers).

The ROC curves illustrated in Fig. 4a and b for ResNet-18 and ResNet-50 models, respectively, help in assessing the CNN models' performance in classification problems at various threshold levels. It is demonstrated that the curves rise sharply and approach towards the top-left corner indicating better classification performance (i.e., TPR approaches to 1 and FPR approaches to 0) and high capability of distinguishing between welding defects' classes.

The AUC, the area under the ROC curve, represents the degree of separability of the CNN model. The higher the AUC values, the strong classification ability of the CNN model. Therefore, ResNet-50 model (Fig. 4b) successfully achieved almost perfect classification performance since the AUC score for each weld defect class is above 0.99. ResNet-18 model (Fig. 4a) has slightly lower AUC-ROC scores (e.g., cracks and lack of penetration predicted have 0.98 score).

Shown in Fig. 5a and b is the confusion matrix created for the defect classification task for CNN classifier trained using ResNet-18 and ResNet-50 models, respectively,

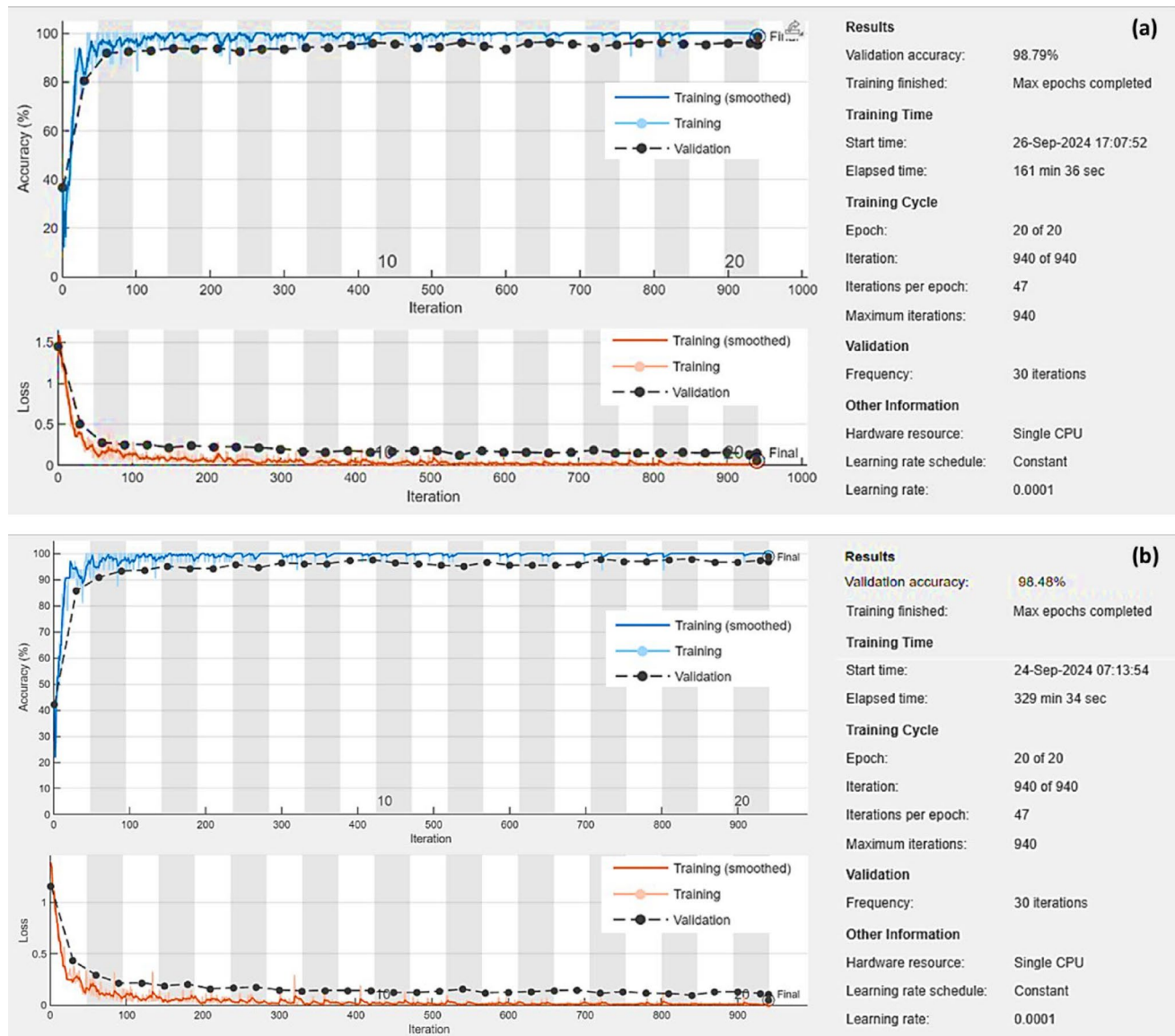


Fig. 3 Train learning and validation curves of CNN classification models; **a** ResNet-18 and **b** ResNet-50

Table 1 Quantitative performance evaluation metrics for CNN architectures used in welding defects classification

CNN Architecture	Weld defect	Quantitative performance metrics			
		Precision	Recall	F1-score	AUC
ResNet-18	Crack	0.96809	0.82727	0.89216	0.98731
	Lack of penetration	0.86777	0.95455	0.90909	0.99959
	Porosity	0.95652	1	0.97778	0.98992
ResNet-50	Crack	1	0.90909	0.95238	0.99715
	Lack of penetration	0.96491	1	0.98214	0.99492
	Porosity	0.92241	0.97273	0.9469	0.99942

illustrating the total number of correctly classified and misclassified images. It is notable that ResNet-50 model predicts the classes of the images of the test dataset almost

perfectly. Although both ResNet-18 and ResNet-50 models 100% correctly predict the porosity defect class, ResNet-50 fails in predicting only 3 lack of penetration images being

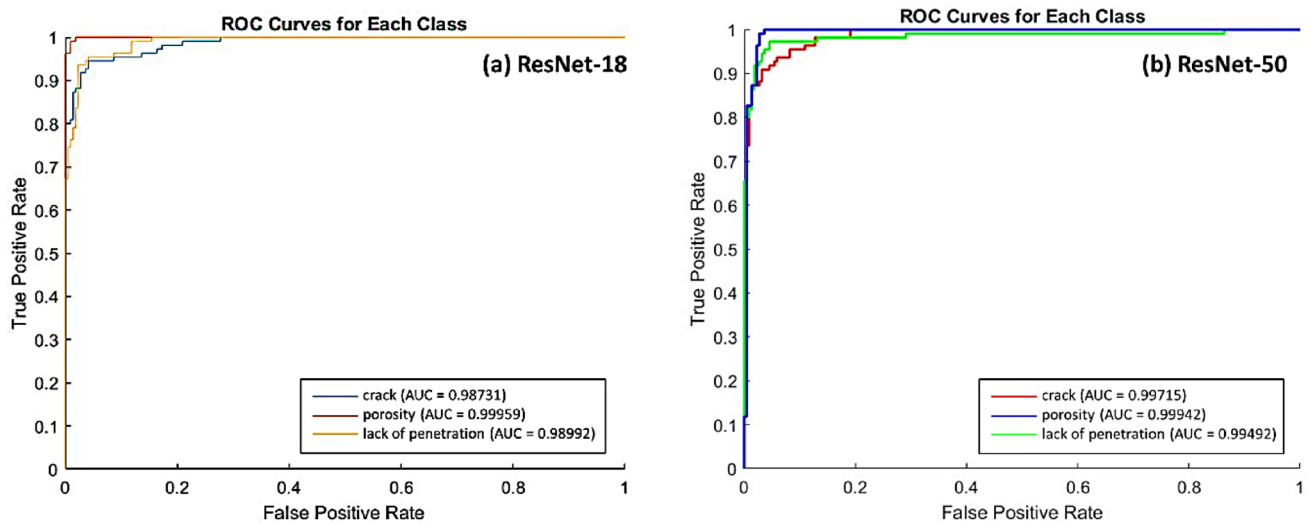


Fig. 4 Receiver Operating Characteristic (ROC) curves for **a** ResNet-18 and **b** ResNet-50 models

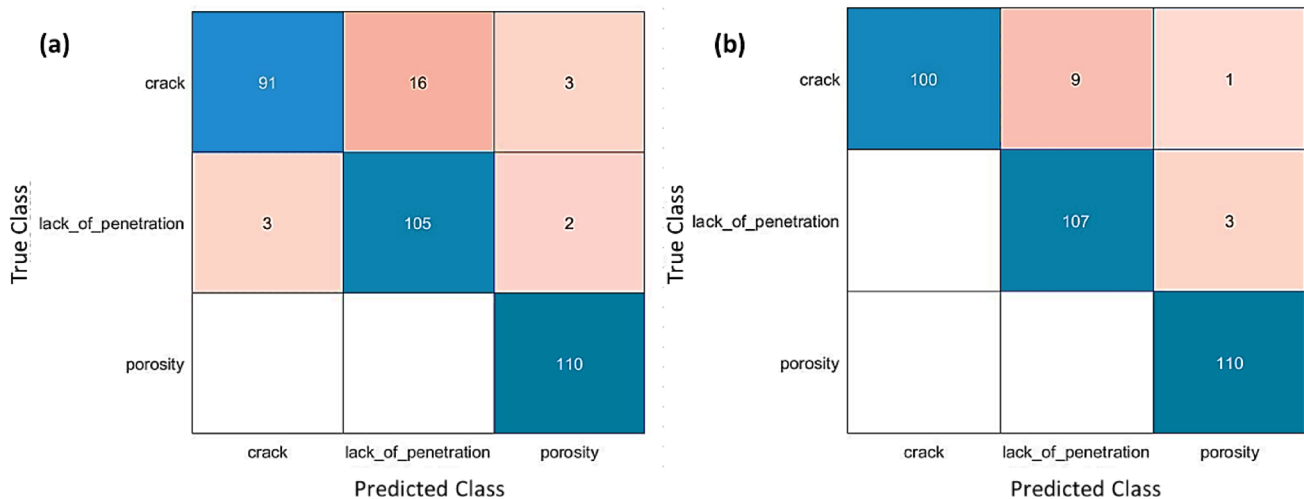


Fig. 5 Confusion matrix created for the defect classification task for CNN classifier trained using **a** ResNet-18 and **b** ResNet-50

incorrectly predicted as porosity defect, while ResNet-18 fails in predicting 5 images of lack of penetration defect and predicts three of them as crack and the other two as porosity defect class. Moreover, ResNet-50 model successfully predicts 100 images of crack defect, with exception to nine images incorrectly predicted as lack of penetration and one image as porosity defect class. However, ResNet-18 model predicts only 91 images out of 110 of crack defect images, and incorrectly predicts 16 images as lack of penetration and 3 images as porosity defect class.

The Grad-CAM algorithm has been implemented to understand the CNN classification process, enabling hypotheses about the potential causes of misclassification. A comparison of Grad-CAM heatmaps visualization

of each weld defect class is shown in Fig. 6a and b using ResNet-18 and ResNet-50 modelling, respectively. The Grad-CAM heatmaps offer high clarity of information which reveal that the used ResNet models are focusing on the features for the radiographic images of defective weld classes. This indicates that the model fits the training data well and the classification images dataset falls in acceptable range. By visual inspection of the figures, it can be seen that both ResNet-18 and ResNet-50 models identified the weld porosity characteristics successfully. The Grad-CAM heatmaps explanation shows that both ResNet models are focusing on similar parts that coincide with the corresponding radiographic image of porosity defect.

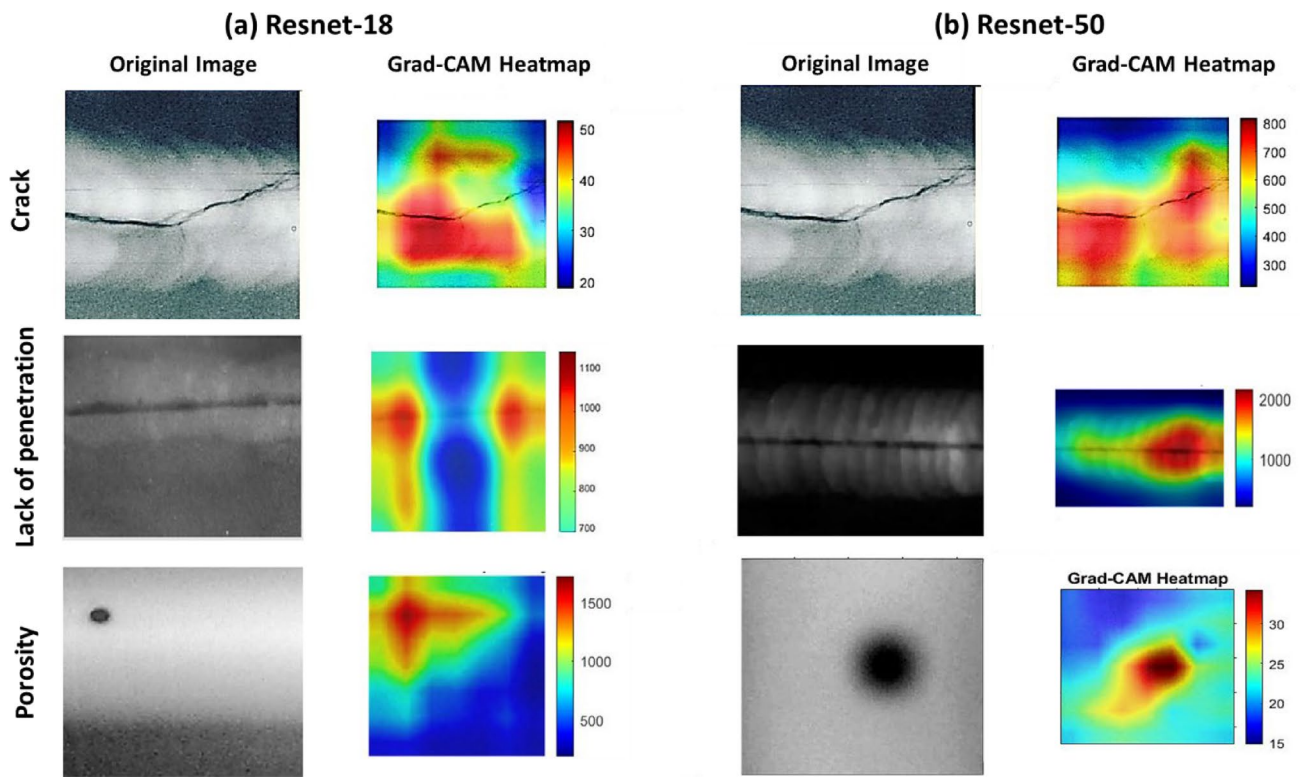


Fig. 6 Comparison of Grad-CAM heatmaps using **a** ResNet-18 and **b** ResNet-50 modelling

However, Grad-CAM heatmaps of crack and lack of penetration defects using ResNet-18 model are less helpful as it provides inaccurate images than those of ResNet-50 model. In certain instances, ResNet-18 model fails to detect some features related to the weld defect, while as ResNet-50 model has a strongest response in the weld defects regions.

4 Conclusions

This paper presented an effective approach for welding defects detection and classification based on the integration of CNNs with machine vision systems. A CNN-based machine vision system was employed to detect and classify three types of weld defects in radiographic images (crack, lack of penetration, and porosity). Specifically, two transfer learning algorithms; ResNet-18 and ResNet-50; were applied and assessed to determine the most effective method.

For our proposed CNN classification models, it was shown by the comparative results that the welding defects were not only effectively detected with high accuracy, but also the defects types were classified accurately. The average test accuracy for ResNet-18 and ResNet-50 models was 92.727% and 96.061%, respectively. However, ResNet-50 classification model was the best in terms of

precision, recall, and F1-score in identifying the defective instances (100%, 90.9%, and 95.23% for crack and 96.49%, 100%, and 98.21% for lack of penetration defects, respectively). Also, ResNet-50 model successfully achieved almost perfect classification performance in terms of AUC score, confusion matrix, and Grad-CAM heatmap results.

Summing up, by leveraging the power of deep learning, the study aims to improve defect classification accuracy, streamline quality control processes, and pave the way for intelligent, automated welding inspection systems. Thus, our proposed system can be very powerful alternative for sectors that apply automated welding techniques, as it significantly saves manpower and costs and avoids defective products.

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Declarations

Conflict of interest The authors have no conflicts of interest to disclose.

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